

Transfer Learning with ResNet-50 for Detecting AI-Generated Images

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Abstract

Text-to-image generative models such as DALL·E now produce synthetic images that are difficult to separate from genuine photographs, raising practical concerns for misinformation and media authenticity. In an earlier study we trained a compact three-block convolutional neural network (SIMPLECNN) from scratch for the binary task of real-versus-AI image classification and reached only about 78% test accuracy, attributing the ceiling to an over-parameterised dense classifier, a small training set, and the absence of pretrained features. In this work we realise the transfer-learning roadmap proposed there. We fine-tune a ResNet-50 pretrained on ImageNet, replacing only its final classification layer, and train it on a balanced 7,561-image subset of the DALL·E Recognition Dataset using a proper three-way train/validation/test split, stronger augmentation, the AdamW optimiser with a cosine learning-rate schedule, and best-model checkpointing. The fine-tuned network attains 93.30% test accuracy with an F1-score of 0.934, a gain of roughly fifteen percentage points over the from-scratch baseline while training for only four epochs. We analyse the model through learning curves, a confusion matrix, and qualitative predictions, explain the residual-learning mechanism that makes such rapid convergence possible, and identify the remaining sources of error and a path toward further gains.

1 Introduction

Modern generative models synthesise photorealistic images directly from text prompts, and systems such as DALL·E have placed this capability in the hands of a broad public. As the visual fidelity of synthetic images approaches that of real photographs, automatically distinguishing “real” from “AI-generated” content has become important for combating misinformation, protecting media provenance, and preserving trust in visual evidence.

Convolutional neural networks (CNNs) are the standard tool for image classification and have driven the field since deep CNNs decisively outperformed classical methods on large-scale recognition benchmarks [1]. A recurring lesson from that literature is that the dominant cost of training a deep network is *learning good visual features*, and that this cost can be largely avoided by reusing features learned on a large dataset—the principle of *transfer learning*.

This paper is a direct follow-up to our own from-scratch baseline. In the prior study we deliberately trained a lightweight three-block CNN, SIMPLECNN, with no pretrained weights, and observed a test accuracy of approximately 78%. The analysis there isolated three causes of the ceiling: roughly 99.6% of the model’s parameters sat in a single dense layer that behaved as an over-fitting magnet; the network was trained from scratch on only a few thousand images; and it omitted normalisation and best-model selection. We argued that the single highest-leverage change would be transfer learning with a pretrained residual network. The present work implements exactly that change and quantifies its effect.

Our contributions are as follows. *First*, we replace the from-scratch CNN with a ResNet-50 pretrained on ImageNet, fine-tuning only a lightweight two-component classification head, and we explain the residual-learning mechanism that lets the network converge in a handful of epochs. *Second*, we correct the methodological weaknesses of the baseline by adopting a stratified three-way

split, stronger label-preserving augmentation, the AdamW optimiser with a cosine schedule, and

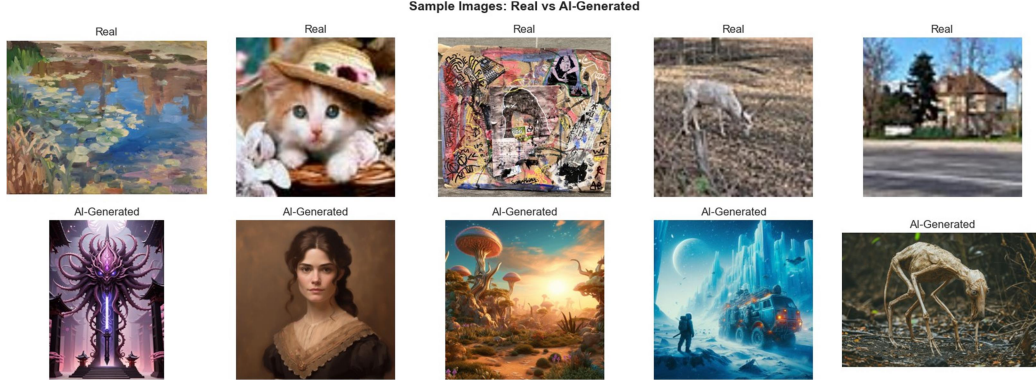


Figure 1: Representative images from the two classes. Top: real photographs (label 0). Bottom: AI-generated images (label 1). The synthetic class is visually diverse, which both helps generalisation and makes a handful of borderline cases genuinely hard.

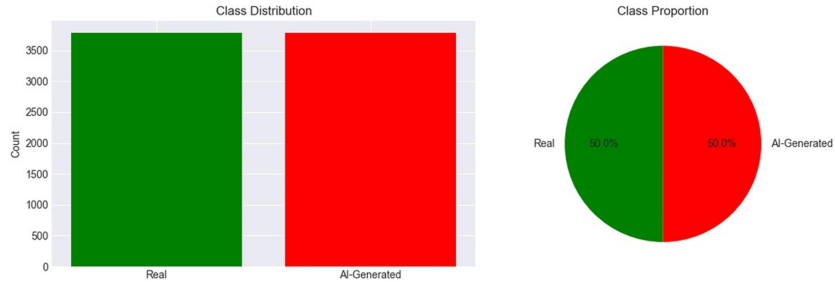


Figure 2: Class distribution of the balanced sample: 3,780 real and 3,781 AI-generated images. Balanced evaluation makes accuracy a meaningful summary statistic.

Checkpointing on the best validation accuracy. *Third*, we provide a careful empirical evaluation—learning curves, per-class metrics, a confusion matrix, and qualitative predictions including a failure case—and a critical discussion of the remaining error sources and the risk of dataset artefacts.

2 The Dataset

We use the DALL·E Recognition Dataset [8], a publicly available collection of real photographs and AI-generated images. The full dataset contains 3,781 real images (label 0) and 17,857 AI-generated images (label 1), a class imbalance of roughly 1:4.7 in favour of the synthetic class. Figure 1 shows representative examples from each class; the synthetic images span a wide stylistic range, from photorealistic portraits to clearly fantastical scenes, while several “real” images are themselves paintings or heavily edited photographs—an ambiguity we return to in Section 5.

Balanced sampling. Training directly on the imbalanced set would bias the classifier toward the majority synthetic class. As in the baseline, we draw a *balanced* subset by randomly sampling an equal number of images from each class with a fixed seed for reproducibility. Because the real folder caps the balanced size, we sample 3,780 real and 3,781 AI-generated images, giving 7,561 images in total. This is roughly $3.8\times$ larger than the 2,000-image subset used for the SIMPLECNN baseline, directly addressing the second weakness identified there. Figure 2 confirms the near-perfect class balance of the sampled set. We deliberately avoid pixel-space oversampling techniques such as SMOTE, which interpolate between feature vectors and produce blurred, semantically meaningless images when applied to raw pixels. As a safety net we additionally compute balanced class weights for the loss; on the balanced sample these weights are both 1.000, but they would rebalance the objective automatically if the full imbalanced set were used.

Table 1: Stratified 70/15/15 split of the balanced sample. Class balance is preserved in every partition.

Partition	Real (0)	AI (1)	Total
Train	2,646	2,646	5,292
Validation	567	567	1,134
Test	567	568	1,135
Total	3,780	3,781	7,561

Three-way split. A methodological flaw of the baseline was that the test set was used both to monitor training and to report final numbers, so its figures were effectively validation performance. We fix this with a stratified 70/15/15 split into train, validation, and test partitions. The test partition is touched only once, after model selection is complete. Table 1 lists the resulting sizes; class balance is preserved in every partition.

Pre-processing and augmentation. All images are converted to RGB, resized to 224×224 , and normalised with the standard ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$), matching the distribution ResNet-50 was pretrained on. Training augmentation is deliberately stronger than the baseline’s flip-and-rotate scheme: we apply `RandomResizedCrop` (scale 0.7–1.0), random horizontal flips, `Color5itter` (brightness, contrast, and saturation each ± 0.2), and small random rotations up to $\pm 10^\circ$. Random-resized cropping forces the model to cope with different framings, while colour jitter prevents it from latching onto a particular lighting signature. Evaluation uses a deterministic resize with no augmentation.

3 Method

3.1 Why transfer learning?

Training a CNN from scratch on a few thousand images means starting from random weights and hoping the network discovers useful visual features within a few epochs—precisely the regime in which the SIMPLECNN baseline struggled. ResNet-50 [2] instead arrives already knowing how to detect edges, textures, materials, shapes, and thousands of higher-level patterns, because it was pretrained on ImageNet’s 1.2 million labelled images. Fine-tuning reuses that learned visual knowledge and asks the network to learn only the final “is this real or AI?” decision, which is a far smaller learning problem and explains the rapid convergence reported in Section 4.

3.2 Residual learning: behind the black box

The reason a 50-layer network can be trained at all is the *residual block*, the core idea of ResNet. In a conventional deep stack, each group of layers must learn a full transformation $H(x)$ of its input, and stacking many such groups makes optimisation hard: gradients vanish and very deep plain networks paradoxically perform *worse* than shallow ones. A residual block reframes the problem. Rather than learning $H(x)$ directly, the two weight layers inside the block learn only the *residual* $F(x) = H(x) - x$, and the block’s output is formed by adding the untouched input back through an identity “skip” connection:

$$y = F(x) + x = W_2 \sigma W_1 x + x, \quad (1)$$

where σ is the ReLU nonlinearity and W_1 , W_2 denote the (convolutional) weight layers. Figure 3 illustrates this structure. The skip connection has two crucial effects. First, it gives gradients a direct, unobstructed path back to earlier layers during back-propagation, which keeps very deep networks trainable. Second, it makes the identity mapping trivial to represent—if a block is not useful, it can simply drive $F(x) \rightarrow 0$ and pass its input through unchanged—so adding depth never hurts and often helps. ResNet-50 stacks 16 such bottleneck blocks; this is what lets it act as the powerful, pretrained “black box” feature extractor we build on.

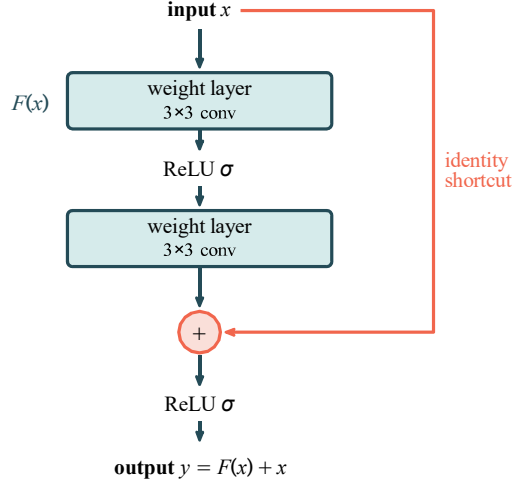


Figure 3: A residual block—the building block of ResNet. The two weight layers learn only the residual $F(x)$, while the identity shortcut (coral) adds the input back. This keeps gradients flowing and makes depth safe.

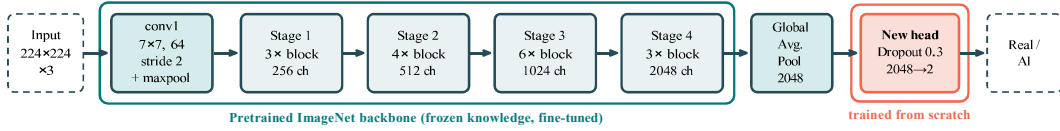


Figure 4: ResNet-50 pipeline used in this work. The ImageNet-pretrained backbone (teal) extracts a 2,048-dimensional embedding via four stages of bottleneck residual blocks and global average pooling; only the final classification head (coral) is replaced and adapted to the two-class real-versus-AI decision. The full model has 23,512,130 parameters.

3.3 Architecture and head replacement

We load ResNet-50 with ImageNet-pretrained weights (INAGENET1K_V2). The network begins with a 7×7 convolution and max-pooling stem, followed by four stages of bottleneck residual blocks (3, 4, 6, and 3 blocks respectively) that progressively halve the spatial resolution while widening the channel dimension from 256 to 2,048. A global average-pooling layer then collapses each feature map to a single value, yielding a 2,048-dimensional embedding. In the original network this embedding feeds a $2,048 \rightarrow 1,000$ layer for the thousand ImageNet classes; we replace it with a small head suited to our binary problem—a dropout layer (rate 0.3) followed by a $2,048 \rightarrow 2$ linear layer:

```
model.fc = Sequential(Dropout(0.3), Linear(2048, 2)).
```

Figure 4 shows the full pipeline with the replaced head highlighted. The complete model has 23,512,130 parameters. Crucially, in contrast to SIMPLECNN—where a single dense layer held 99.6% of the weights—ResNet-50 replaces the parameter-heavy flatten-and-dense classifier with global average pooling, so capacity is distributed across the convolutional hierarchy rather than concentrated in one over-fitting-prone layer.

3.4 Training setup

The network is fine-tuned by minimising a class-weighted cross-entropy loss with the AdamW optimiser. Four choices distinguish this setup from the baseline and reflect the fact that we are adapting valuable pretrained weights rather than learning from scratch:

- **A low learning rate** of 10^{-4} (versus 10^{-3} for the baseline). Pretrained weights are valuable; a high rate would overwrite them. We make small, careful updates.

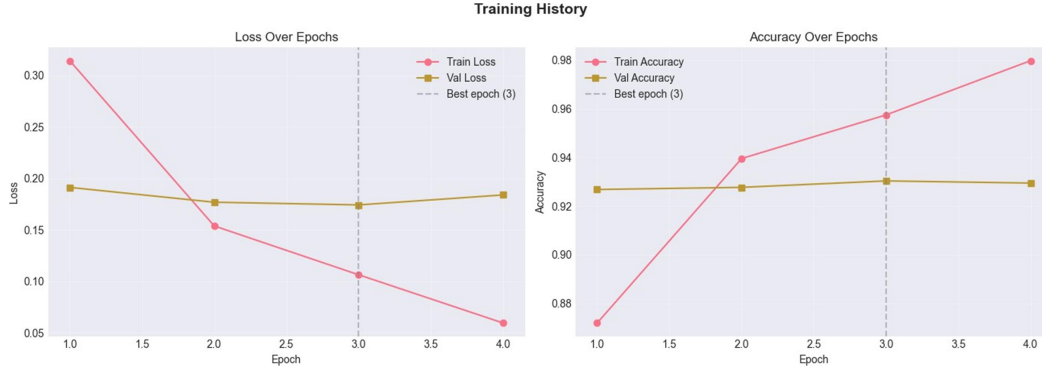


Figure 5: Training history over four epochs. Left: loss. Right: accuracy. Training metrics (pink) improve monotonically, but validation metrics (gold) plateau after epoch 3 (dashed line), with validation loss rising at epoch 4: the signature of incipient over-fitting. Best-model checkpointing keeps the epoch-3 weights.

- **AdamW with weight decay 10^{-4}** , which decouples regularisation from the gradient update and generalises slightly better than plain Adam.
- **A cosine-annealing schedule**, which decays the learning rate smoothly to zero over training instead of the baseline’s abrupt step halvings.
- **Best-model checkpointing**: validation accuracy is recorded after every epoch and the weights are saved whenever it improves, so the reported model is the best epoch rather than the last.

We use a batch size of 32 and train for only four epochs—more is unnecessary, as Section 4 shows. Training was performed on an Apple-Silicon GPU through PyTorch’s Metal (MPS) backend; the code falls back automatically to CUDA or CPU. All randomness is seeded for reproducibility.

4 Experiments and Results

Training dynamics. Figure 5 shows the learning curves. The model is already strong after a single epoch, reaching 92.68% validation accuracy, which confirms the value of the pretrained features: the network does not have to learn vision from scratch, only to specialise. Validation accuracy peaks at 93.03% at epoch 3, and best-model checkpointing retains exactly those weights. The fourth epoch is instructive: training accuracy continues climbing to 97.98% while validation accuracy dips slightly to 92.95% and validation loss ticks up—the onset of mild over-fitting. This is precisely the situation checkpointing is designed to guard against, and it vindicates the decision to keep the epoch-3 model rather than the final one.

Test-set performance. We evaluate the checkpointed model once on the untouched test partition. The results appear in Table 2. The fine-tuned ResNet-50 attains 93.30% accuracy with precision, recall, and F1-score all close to 0.93–0.94. Because the test set is balanced, accuracy is a faithful summary, and the tight agreement among the four metrics indicates the classifier is not trading one type of error for another.

Confusion matrix and per-class behaviour. Figure 6 reports the confusion matrix. Of 567 real images, 525 (92.59%) are classified correctly; of 568 AI-generated images, 534 (94.01%) are correct. The two error types are close in magnitude—42 real images mistaken for AI and 34 AI images mistaken for real—so the model exhibits no strong directional bias. The slightly higher recall on the synthetic class is consistent with AI-generated images often carrying subtle but consistent generative signatures that the network can latch onto.

Qualitative predictions. Figure 7 shows predictions on six random test images together with the model’s confidence. Most are correct with very high confidence (1.00), including a stylised cartoon character and a pencil-style portrait. The informative case is the bottom-centre image: an AI-generated parasol that the model labels “Real” with a low confidence of 0.62. Its clean studio

Table 2: Classification performance on the untouched test set (1,135 images). Treating “AI-generated” as the positive class.

Metric	Value
Accuracy	0.9330
Precision	0.9271
Recall	0.9401
F1-score	0.9336
Best validation accuracy (epoch 3)	0.9303

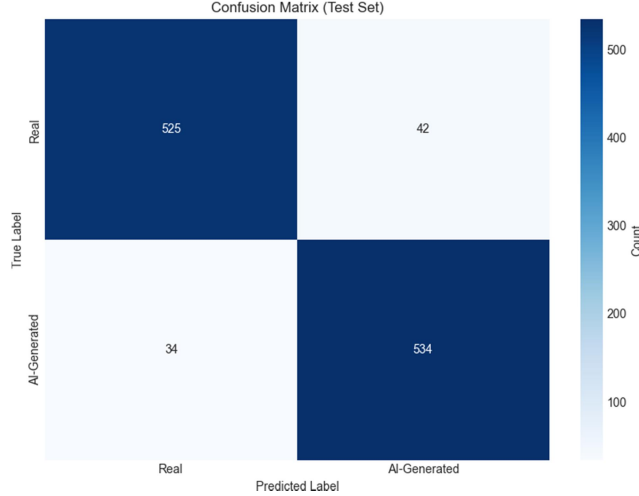


Figure 6: Confusion matrix on the test set. Errors are nearly symmetric between the two classes (42 versus 34), indicating a well-calibrated decision boundary rather than a biased one.

lighting and plausible geometry place it near the decision boundary, and the low confidence shows the network is appropriately uncertain rather than confidently wrong. Such borderline, photorealistic synthetic objects are exactly where detection remains hardest.

Comparison with the from-scratch baseline. Table 3 situates the fine-tuned model against the SIMPLeCNN baseline. Transfer learning lifts test accuracy from roughly 78% to 93.3%—a gain of about fifteen percentage points—despite training for only four epochs rather than twenty. The improvement is the combined effect of pretrained features, a larger and properly split dataset, stronger augmentation, and best-model selection, but the dominant factor is the reuse of ImageNet features, exactly as the baseline study predicted.

5 Discussion

The results confirm the central hypothesis of the baseline study: on a small dataset, reusing pre-trained features is decisively more effective than learning them from scratch. Three points deserve a more critical reading.

Where the remaining errors come from. The 6.7% of misclassified images are not uniformly distributed. As the qualitative analysis suggests, the hard cases cluster at the boundary between the two classes’ visual distributions: photorealistic synthetic objects with plausible lighting and geometry, and “real” images that are themselves paintings, sketches, or heavily stylised photographs (several appear in Figure 1). In other words, part of the residual error reflects genuine label ambiguity in the dataset rather than model weakness, and no amount of additional training will drive it to zero.

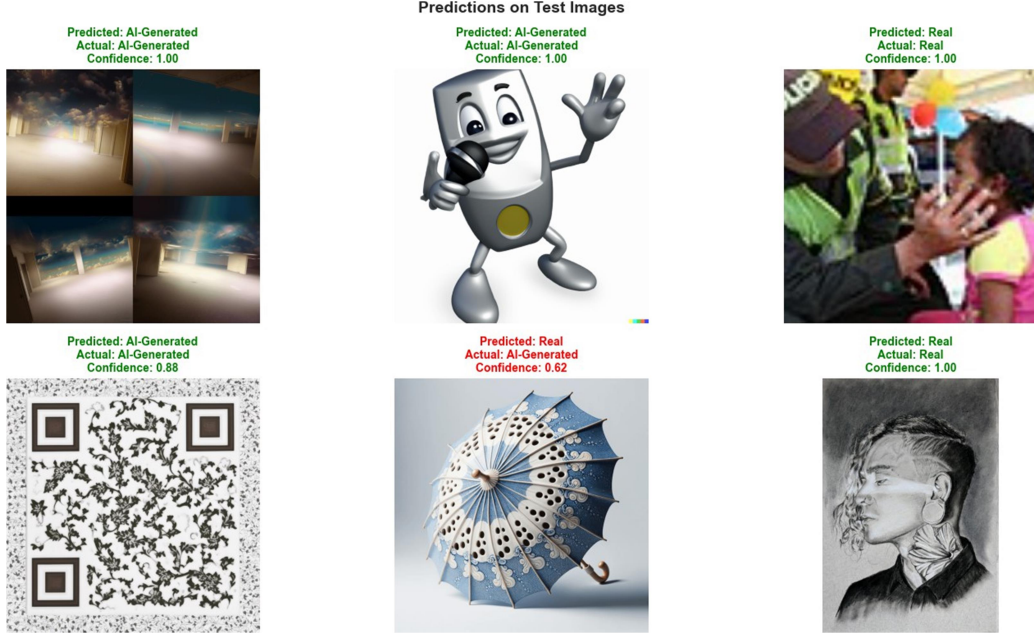


Figure 7: Predictions on six random test images. Green titles denote correct predictions, red incorrect. Confidence scores are shown beneath each image. The single error (bottom centre, an AI-generated parasol predicted “Real”) is a low-confidence borderline case, illustrating where photorealistic synthetic content still defeats the detector.

Table 3: Fine-tuned ResNet-50 versus the from-scratch SIMPLECNN baseline. The ResNet row is a measured result of this work; the SIMPLECNN row is reproduced from the baseline study.

Model	Pretrained	Epochs	Test acc.
SIMPLECNN (baseline)	no	20	$\approx 78\%$
ResNet-50 (this work)	yes	4	93.30%

Mild over-fitting and the value of checkpointing. The divergence between training and validation curves at epoch 4 (Figure 5) shows that even a well-regularised fine-tuning run begins to over-fit quickly once the easy gains are exhausted. With only four epochs the effect is small, but it would compound over a longer schedule. Best-model checkpointing converts this from a liability into a non-issue, and the plateau in validation accuracy after epoch 3 indicates that further epochs would yield little benefit on this dataset.

The dataset-artefact caveat. A caveat specific to AI-image detection carries over from the baseline and is worth restating, because higher accuracy makes it easier to overlook. Detectors can reach deceptively high scores by exploiting incidental dataset artefacts—consistent resolution, compression signatures, or a particular post-processing pipeline—rather than by learning genuine generative fingerprints. A model that scores 93% on this dataset may not transfer to images from a different generator or capture pipeline. Reported accuracies should therefore be read as in-distribution performance, and robustness to unseen generators must be measured separately before any detector is trusted in the wild.

6 Future Work

Several extensions follow naturally from the analysis. (i) *Scale and resolution:* training on the full set of available images, with class weights to handle the natural imbalance, and at a higher input resolution (320 or 384 pixels) would likely add a few points and expose finer generative artefacts.

(ii) *Two-stage fine-tuning*: freezing the backbone to train the head first, then unfreezing for a low-rate full fine-tune, often stabilises adaptation and squeezes out additional accuracy. (iii) *Stronger backbones*: EfficientNet-B0 or ConvNeXt-Tiny offer comparable or better accuracy at lower parameter counts. (iv) *Interpretability*: adding Grad-CAM would reveal which image regions drive each decision and help confirm whether the model attends to genuine generative cues or to dataset artefacts—directly addressing the caveat above. (v) *Cross-generator evaluation*: testing on images from generators not seen during training is the single most important step toward a detector that is useful beyond a single benchmark. A complementary direction is frequency-domain analysis, where many generative models leave periodic artefacts that are hard to perceive in the spatial domain.

7 Conclusion

We presented a transfer-learning approach to detecting AI-generated images that realises the improvement roadmap proposed by our earlier from-scratch baseline. By fine-tuning an ImageNet-pretrained ResNet-50—replacing only a lightweight classification head and correcting the baseline’s methodological weaknesses with a three-way split, stronger augmentation, AdamW with cosine annealing, and best-model checkpointing—we raised test accuracy from roughly 78% to 93.30% while training for just four epochs. The analysis attributes this gain primarily to the reuse of pretrained visual features, locates the residual error in genuinely ambiguous borderline images, and cautions that in-distribution accuracy may not imply robustness to unseen generators. The model and protocol together provide a strong, reproducible baseline for AI-generated image detection and a clear template for the value of transfer learning in small-data settings.

References

- [1] A. Krizhevsky, I. Sutskever, and G. E. Hinton. ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2012.
- [2] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [3] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.
- [4] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- [5] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1):1929–1958, 2014.
- [6] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. ImageNet: A large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009.
- [7] A. Ramesh, M. Pavlov, G. Goh, S. Gray, C. Voss, A. Radford, M. Chen, and I. Sutskever. Zero-shot text-to-image generation. In *International Conference on Machine Learning (ICML)*, 2021.
- [8] DALL·E Recognition Dataset. Kaggle. Accessed 2026.